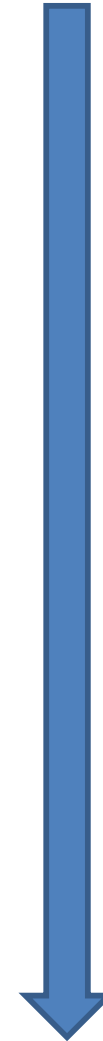


Evolution of Operating System

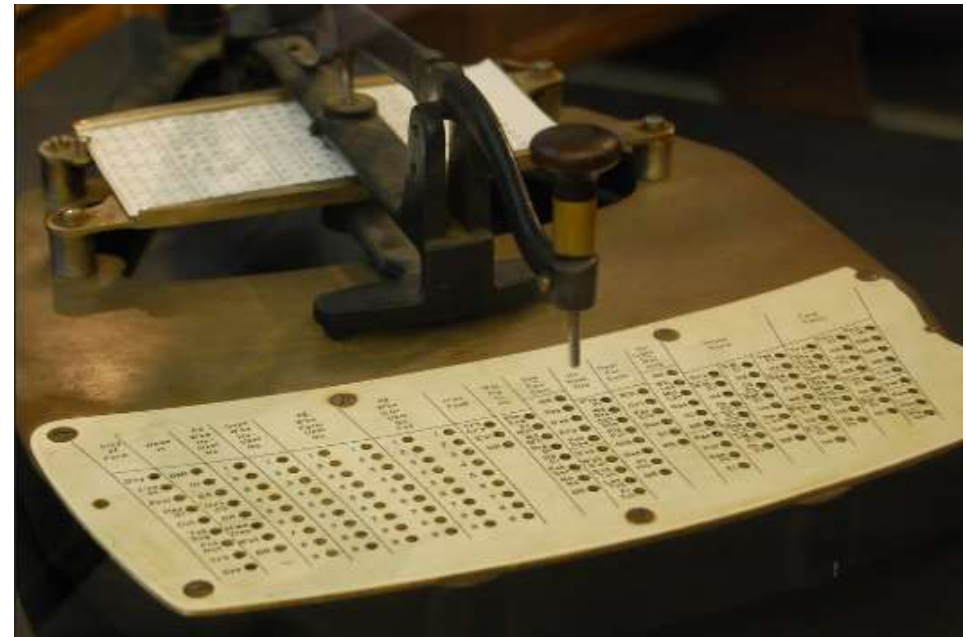
- Mainframe System
 - Batch Systems
 - Multiprogrammed Systems
 - Time Sharing Systems
- Desktop Systems
 - Multiprocessor Systems
 - Symmetric Multiprocessor
 - Asymmetric Multiprocessor
 - Distributed Systems
 - Client Server Systems
 - Peer to Peer Systems
 - Clustered Systems
 - Asymmetric clustering
 - Symmetric clustering
 - Real Systems
 - Hard Real Time System
 - Soft Real Time System
- HandHeld Systems



Mainframe System

- First computers to tackle commercial and scientific applications.
- Growth from Batch systems to Multi-programmed to Timesharing(Multitasking system)

??



??



Batch system

- Early computers were physically enormous machines run from a console.
 - Input devices –
 - Card readers and
 - Tape drivers.
 - O/P devices-
 - Line printers,
 - Tape drivers,
 - Card Punches.

Tape drives

- A tape drive is a data storage device that reads and writes data on a magnetic tape.
 - Magnetic tape data storage is typically used **for offline, archival data storage.**
 - Tape media generally has a **favorable unit cost and a long archival stability.**
- **Magnetic tapes are typically placed in plastic covers referred to as cassettes.**



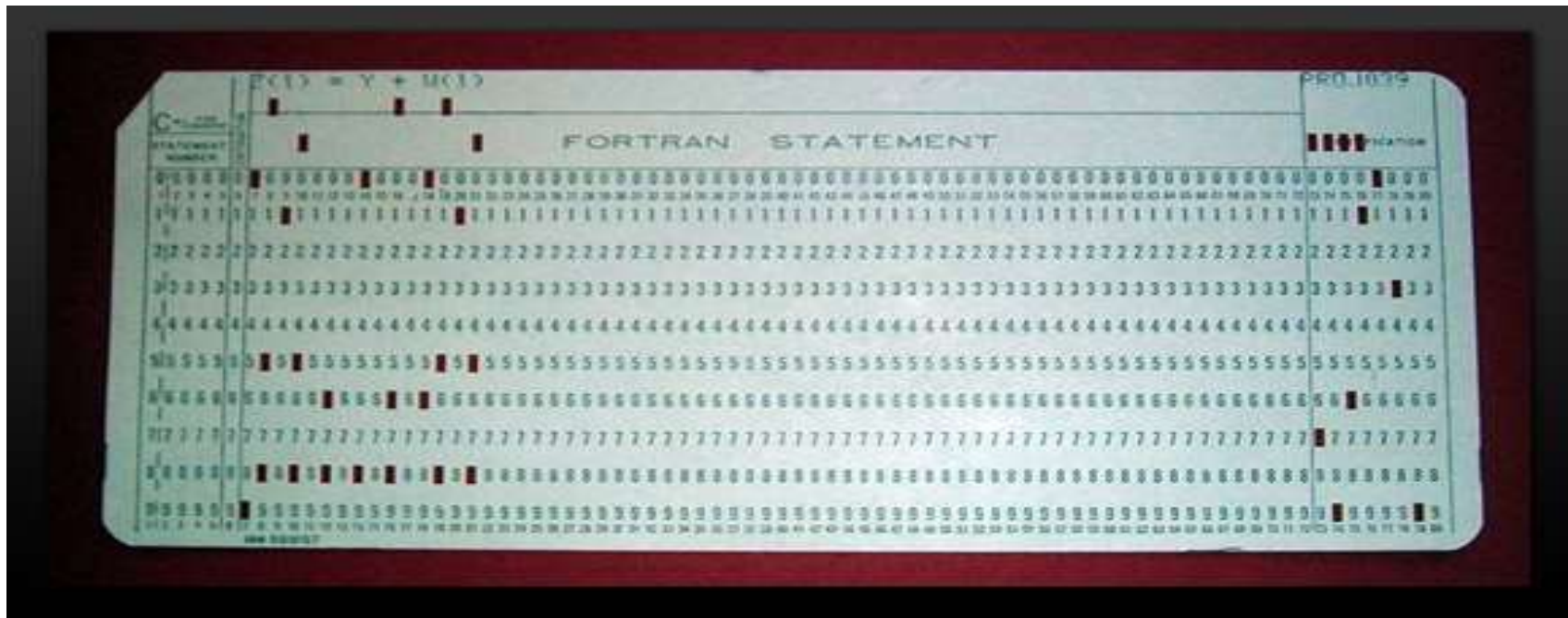
Tape drives

- A tape drive provides sequential access storage, unlike a hard disk drive, which provides direct access storage.
 - A disk drive can move to any position on the disk in a few milliseconds,
 - but a tape drive **must physically wind tape between reels to read any one particular piece of data.**
 - As a result, tape drives have very large average access times.
 - **However, tape drives can stream data very quickly off a tape when the required position has been reached.**

Punch card/A punched card

- Is a piece of stiff paper that can be used to contain digital data
 - represented by the presence or absence of holes in predefined positions.
- Punched cards were widely used through much of the 20th century for
 - data input, output, and storage.

Card from a Fortran program

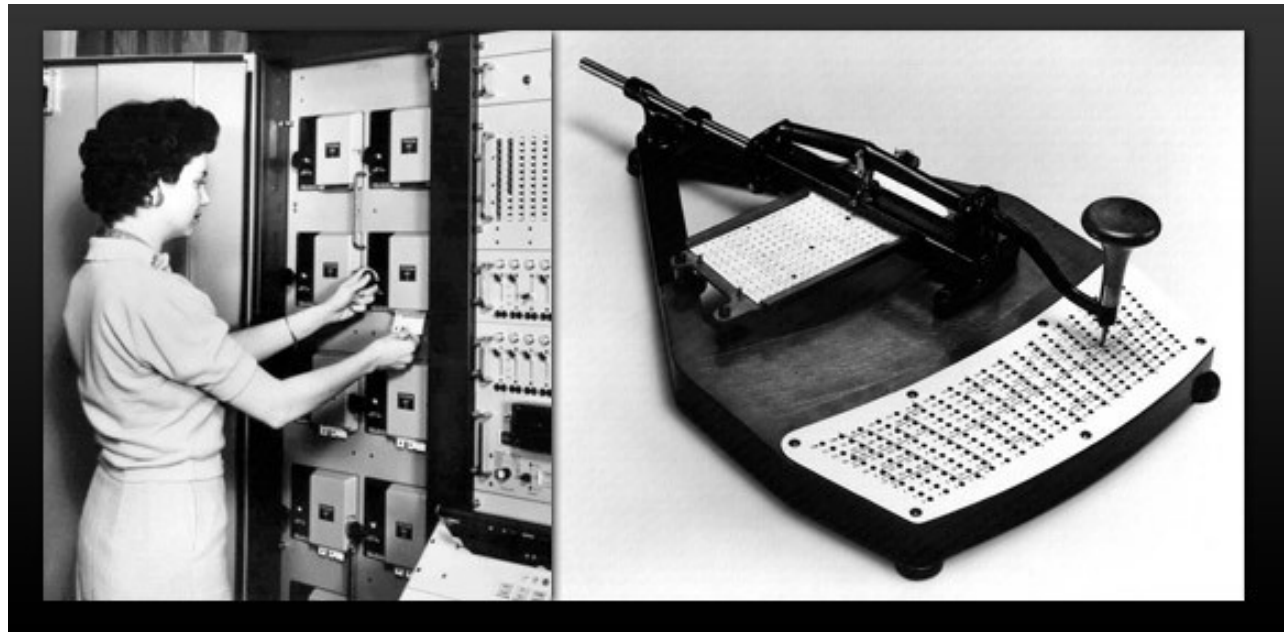


Punch card/A punched card

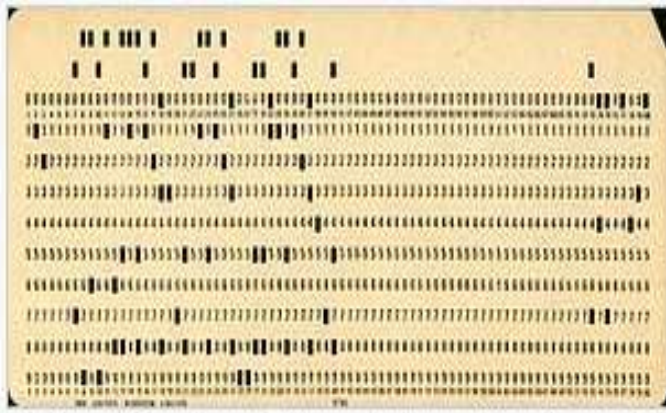
- Many early digital computers used punched cards for input of both
 - computer programs and
 - data.
- **While punched cards are now obsolete as a storage medium, as of 2012, some voting machines still use punched cards to record votes.**

Above left: Punch card reader.

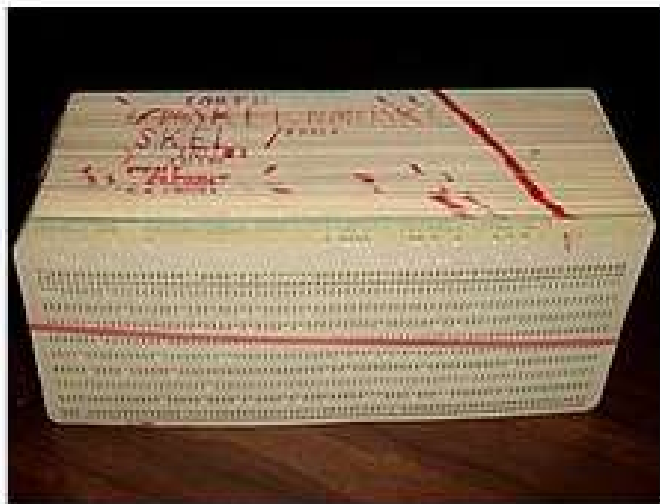
Above right: Punch card writer.



Punch card Reader, A punched card



A 12-row/80-column IBM punched card from the mid-twentieth century



A deck of punched cards comprising a computer program



Card reader/punch

Courtesy :<https://www.ibiblio.org/comphist/node/57>,https://en.wikipedia.org/wiki/Punched_card

Batch system

- User did not interact with the computer systems.
 - User prepared a job which consisted of
 - the programs,
 - data and
 - control information ,
 - submitted it to the computer operator.
 - The job was usually in the form of punch cards.
- At some later time, the output appeared.
 - The o/p consisted of
 - result of the program, as well as
 - a dump of the final memory and
 - register contents for debugging.

Batch system

- To speed up processing,
 - operators batched together jobs with similar needs and
 - ran them throughout the computer as a group.

Batch system

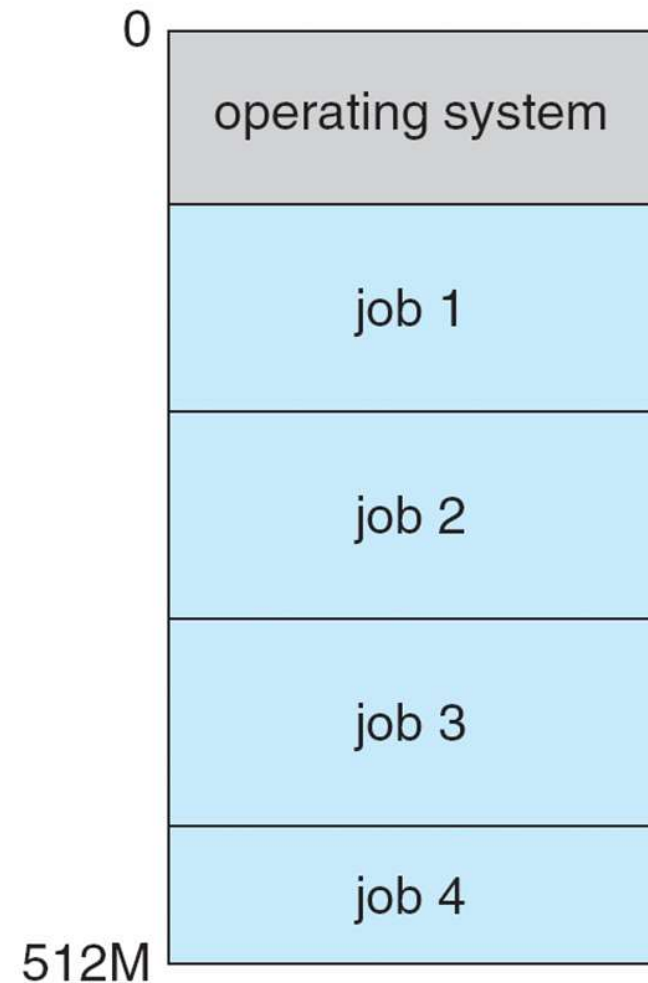
- 1) The programmers would leave their programs with the operator.
- 2) The operator would
 - 1) sort programs into batches with similar requirements and
 - 2) would run each batch as the computer became available.
- 3) The O/P would be sent back to the appropriate programmer

Memory Layout for Multiprogrammed System

- **Bring multiple programs in RAM**

Now,

- **How to execute these programs?**



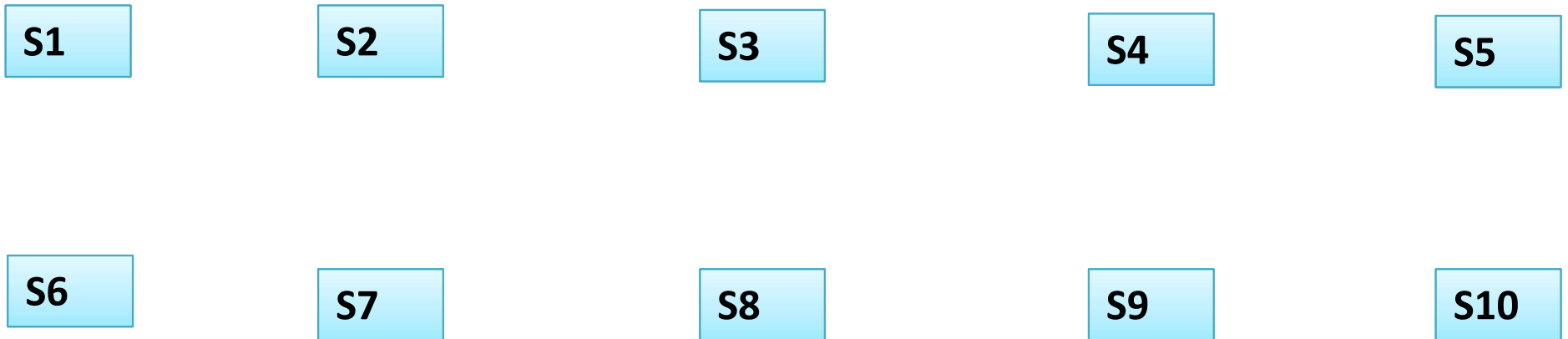
Multiprogramming

Multi programmed –Non-Preemptive?

Multitasking-Preemptive?

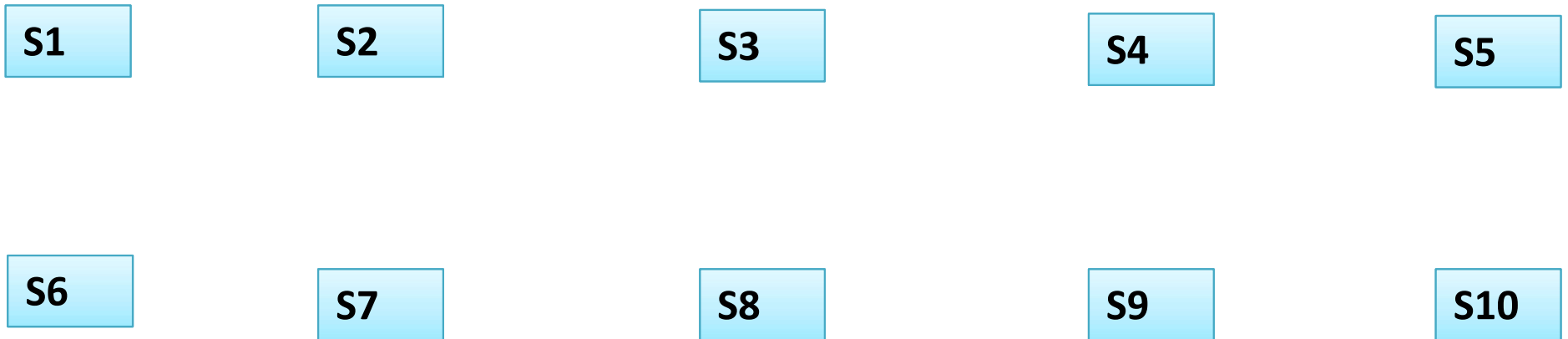
Multiprogramming

Class-Student Example?



Multitasking

Class-Student Example?



Non-multi programmed system

Non-multi programmed system's working –

- As soon as one job leaves the CPU and goes for I/O task the CPU becomes idle.
- **The CPU keeps waiting and waiting until this job (which was executing earlier) comes back and resumes its execution with the CPU.** So CPU remains free for all this while.
- Now it has a **drawback that the CPU remains idle for a very long period of time.** Also, other jobs which are waiting to be executed might not get a chance to execute because the CPU is still allocated to the earlier job.

Multiprogramming

Multi programmed system's working –

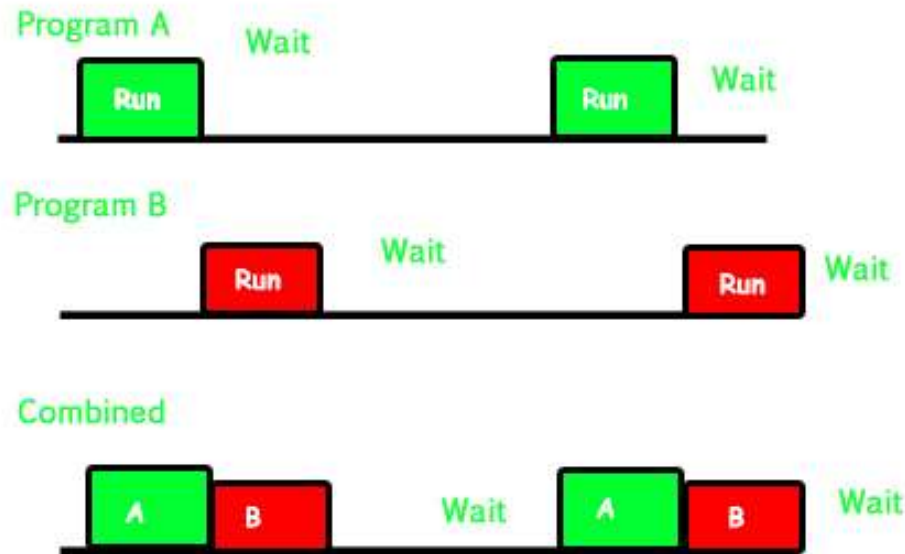
- In a multi-programmed system, as soon as one job goes for an I/O task, the Operating System interrupts that job, chooses another job from the job pool, gives CPU to this new job and starts its execution.
- The previous job keeps doing its I/O operation while this new job does CPU bound tasks.
- Now say the second job also goes for an I/O task, the CPU chooses a third job and starts executing it.

Multiprogramming

- **Multiprogramming** needed for efficiency
 - Single Program cannot keep CPU and I/O devices busy at all times
 - Multiprogramming increases CPU utilization by organizing jobs (code and data) so that CPU always has one to execute.
 - A subset of total jobs in system is kept in memory
 - One job selected and run via **job scheduling**
 - When it has to wait for some task, such as I/O, OS switches to another job
 - In non- multiprogrammed system, The CPU would sit idle.

Multiprogramming

- Program A runs for some time and then goes to waiting state. In the mean time program B begins its execution.



- So the CPU does not waste its resources and gives program B an opportunity to run.

Drawbacks

- Multiprogrammed ,batched system provided an environment
 - where the various system resources were utilized effectively,
 - **but it did not provide for user interaction with the computer system.**

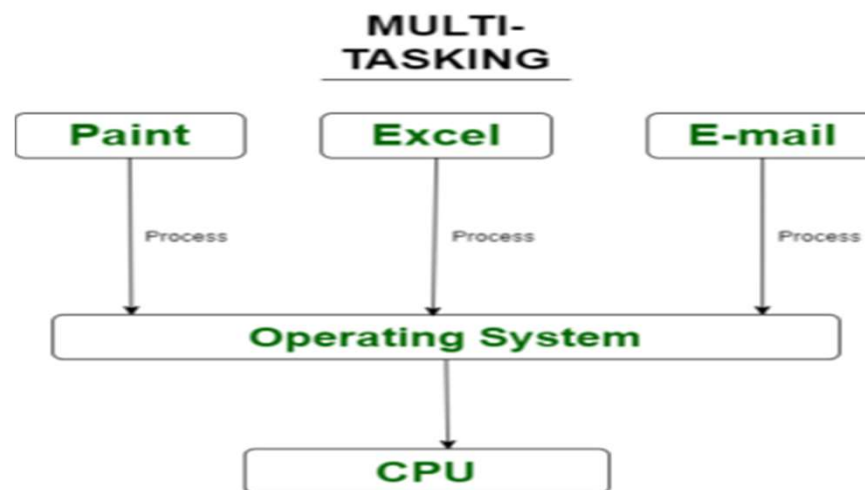
Timesharing (multitasking)

- **Timesharing (multitasking)** is logical extension of multiprogramming
- CPU switches jobs so frequently that users can interact with each job while it is running, creating **interactive** computing.

-Galvin

- CPU executes multiple jobs by switching among them typically using a small time quantum
- The switches occur so frequently that the users can interact with each program while it is running.

- GeeksforGeeks



Timesharing (multitasking)

- An interactive computer system provides direct communication between the user and the system.
- **Response time** should be short , < 1 second
- Each user has at least one program executing in memory
⇒ **process**
- If several jobs ready to run at the same time ⇒ **CPU scheduling**
- If processes don't fit in memory, **swapping** moves them in and out to run
- **Virtual memory** allows execution of processes not completely in memory

Timesharing (multitasking)

- More complex than Multi programmed OS

Timesharing (multitasking)

- To obtain a reasonable response time,
 - jobs may have to be swapped in and out of main memory to the disk.
 - Virtual memory is used
 - which allows execution of a job that may not be completely in memory.
 - Programs can be longer than physical memory.

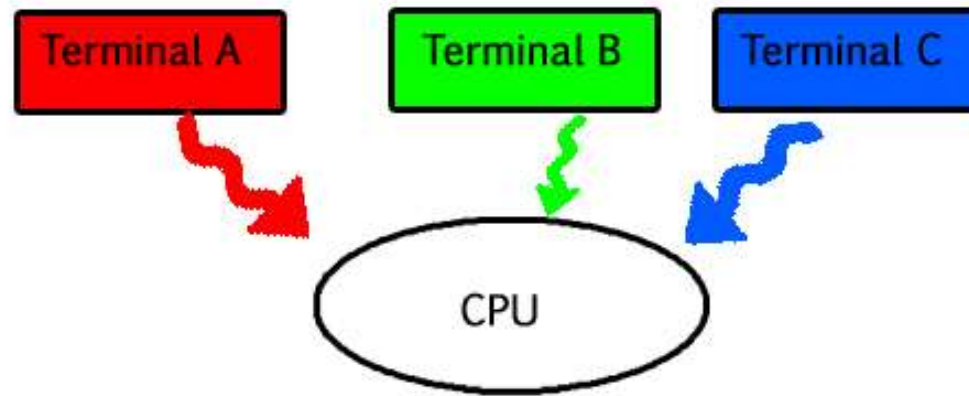
Timesharing (multitasking)

- In a time sharing system, each process is assigned some specific quantum of time for which a process is meant to execute.
- Say there are 4 processes P1, P2, P3, P4 ready to execute.
- Say time quantum of 5 nanoseconds (5 ns).
- As one process begins execution (say P2), it executes for that quantum of time (5 ns).
- After 5 ns the CPU starts the execution of the other process (say P3) for the specified quantum of time.

Timesharing (multitasking)

- Thus the CPU makes the processes to share time slices between them and execute accordingly.
- As soon as time quantum of one process expires, another process begins its execution.
- **The context switch is so fast that the user is able to interact with each program separately while it is running.**
- This way, the user is given the illusion that multiple processes/ tasks are executing simultaneously.

Timesharing (multitasking)



- So for multitasking to take place, firstly there should be multiprogramming i.e. presence of multiple programs ready for execution.
- And secondly the concept of time sharing.
- **Responsiveness Increases**

Multiprogramming Vs Timesharing (multitasking)

Sr.no	Multiprogramming	Multi-tasking
1.	Both of these concepts are for single CPU.	Both of these concepts are for single CPU.
2.	Concept of Context Switching is used.	Concept of Context Switching and Time Sharing is used.
3.	In multiprogrammed system, the operating system simply switches to, and executes, another job when current job needs to wait.	The processor is typically used in time sharing mode. Switching happens when either allowed time expires or where there other reason for current process needs to wait (example process needs to do IO).
4.	Multi-programming increases CPU utilization by organising jobs .	In multi-tasking also increases CPU utilization, it also increases responsiveness.
5.	The idea is to reduce the CPU idle time for as long as possible.	The idea is to further extend the CPU Utilization concept by increasing responsiveness Time Sharing.

ISRO | ISRO CS 2007 | Question 25

What is the name of the technique in which the operating system of a computer executes several programs concurrently by switching back and forth between them?

- (A)** Partitioning
- (B)** Multi-tasking
- (C)** Windowing
- (D)** Paging

ISRO | ISRO CS 2007 | Question 25

What is the name of the technique in which the operating system of a computer executes several programs concurrently by switching back and forth between them?

- (A) Partitioning
- (B) Multi-tasking
- (C) Windowing
- (D) Paging

Answer: (B)

Explanation: In a multitasking system, a computer executes several programs simultaneously by switching them back and forth to increase the user interactivity. Processes share the CPU and execute in an interleaving manner. This allows the user to run more than one program at a time.

GATE | GATE-CS-2002 | Question 46

Which combination of the following features will suffice to characterize an OS as a multi-programmed OS?

- (a) More than one program may be loaded into main memory at the same time for execution.
- (b) If a program waits for certain events such as I/O, another program is immediately scheduled for execution.
- (c) If the execution of program terminates, another program is immediately scheduled for execution.

- (A) a
- (B) a and b
- (C) a and c
- (D) a, b and c

GATE | GATE-CS-2002 | Question 46

Answer: (B)

Which combination of the following features will suffice to characterize an OS as a multi-programmed OS?

Explanation:

- (a) More than one program may be loaded into main memory at the same time for execution.
- (b) If a program waits for certain events such as I/O, another program is immediately scheduled for execution.
- (c) If the execution of program terminates, another program is immediately scheduled for execution.

(a) More than one program may be loaded into main memory at the same time for execution.

True: Only done in a multiprogrammed OS, not in single programmed OS

(b) If a program waits for certain events such as I/O, another program is immediately scheduled for execution.

True: Only done in a multiprogrammed OS, not in single programmed OS

(c) If the execution of program terminates, another program is immediately scheduled for execution.

False: Done in both Multiprogrammed and single programmed OSs

- (A) a
- (B) a and b
- (C) a and c
- (D) a, b and c

Multiprocessor Systems

- Also called **Parallel System or Tightly coupled System** grew in importance.
- **More than one processor in close communication, sharing**
 - the computer bus,
 - the clock and
 - sometimes memory and peripheral devices.

Multiprocessor Systems

Advantages:-

- Increased Throughput
- Economy of scale
- Increased Reliability

Multiprocessor Systems

Advantages:-

- **Increased Throughput**
 - **To Get more work done.**
 - The speedup ratio with N processors is not N , it is less than N .
 - Overhead is incurred in keeping all parts working correctly.
 - Plus Contention for shared resources lowers the expected gains.

Multiprocessor Systems

Advantages:-

- **Economy of scale**
 - **Save more money** as they share
 - peripherals,
 - mass storage,
 - power supplies.

Multiprocessor Systems

Advantages:-

- **Increased Reliability**

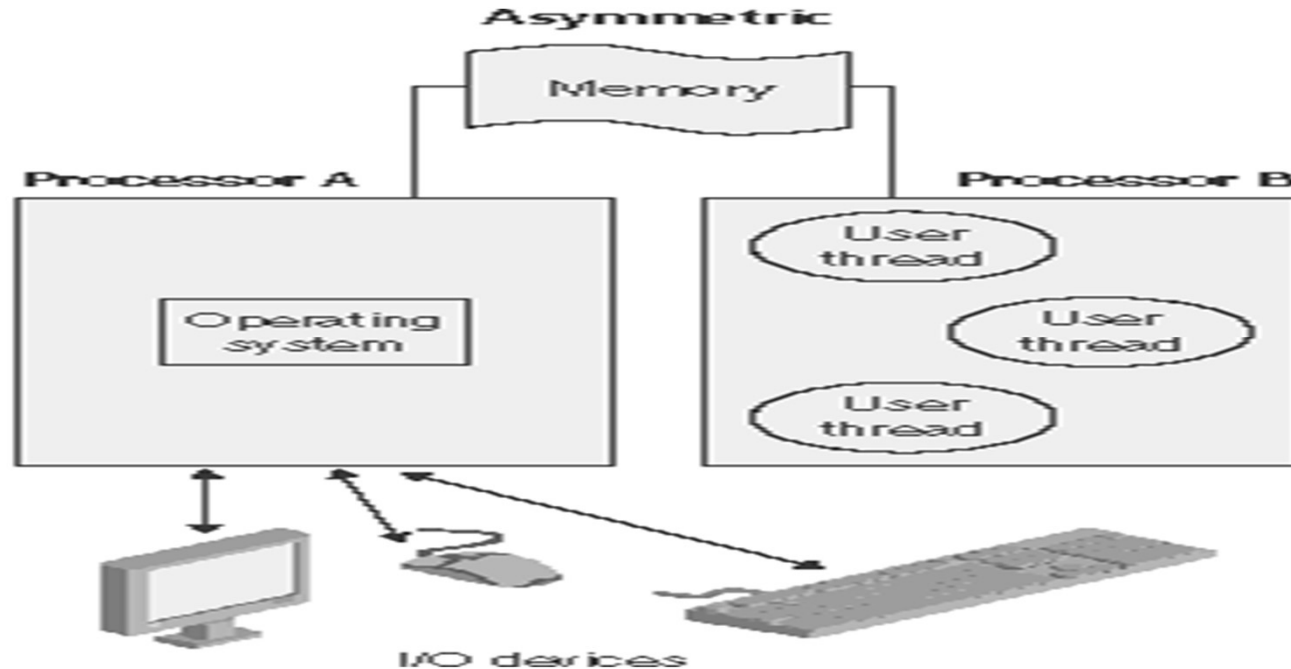
- If functions can be distributed properly among several processors, then
- **the failure of one processor will not halt the system,**
- only slow it down.

Multiprocessor Systems

- Symmetric Multiprocessor
 - Each processor runs an identical copy of the OS and
 - These copies communicate with one another as needed.
- Asymmetric Multiprocessor
 - Each processor is assigned a specific task.
 - A master processor controls the system,
 - The other processor either look to the master for instruction or have predefined tasks.
 - The master allocates the work to the slave processor.

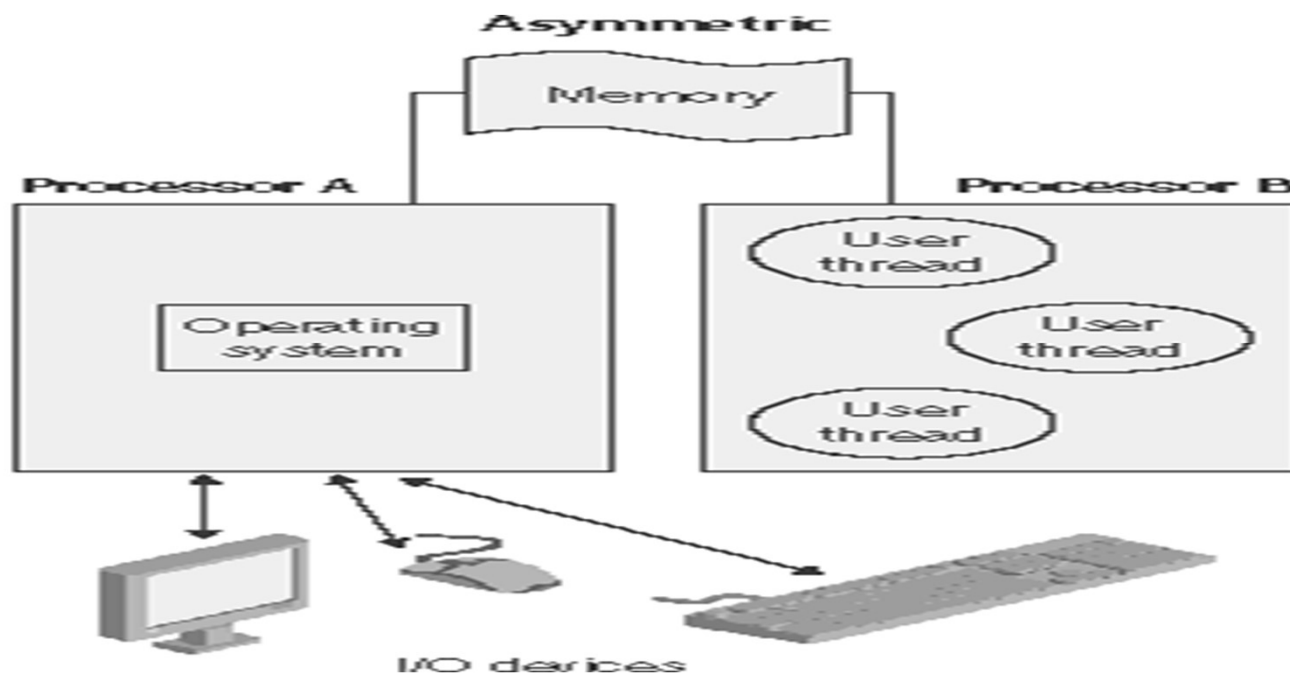
Asymmetric multiprocessing

- ASMP
 - The operating system typically sets aside one or more processors for its exclusive use.
 - The remainder of the processors run user applications.



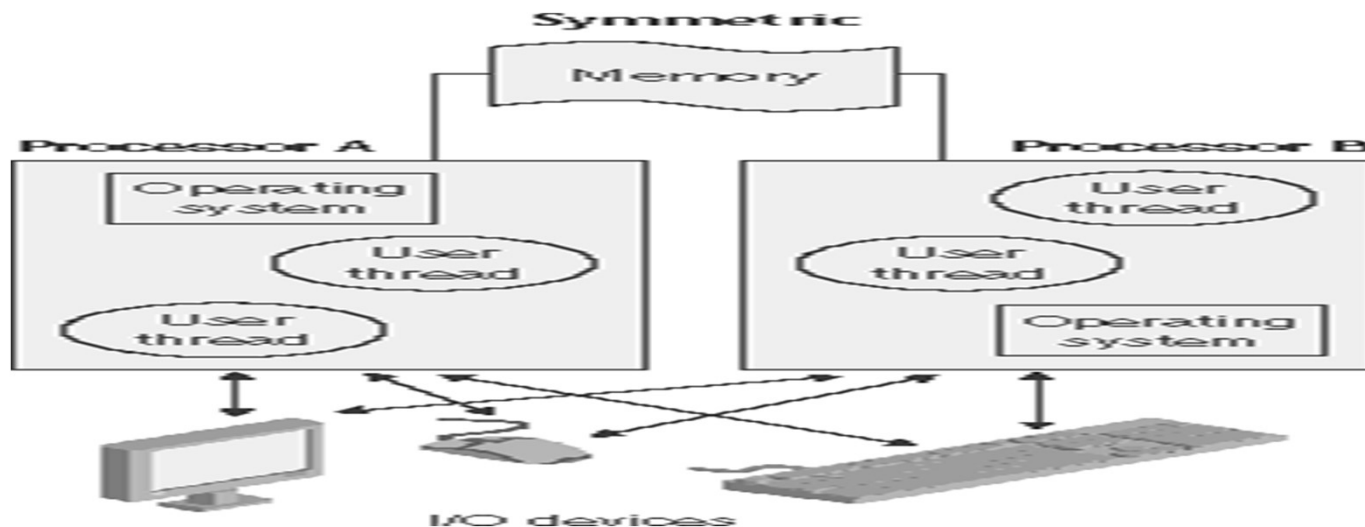
Asymmetric multiprocessing

- In the ASMP model,
 - If the processor that fails is an operating system processor,
 - The whole computer can go down.



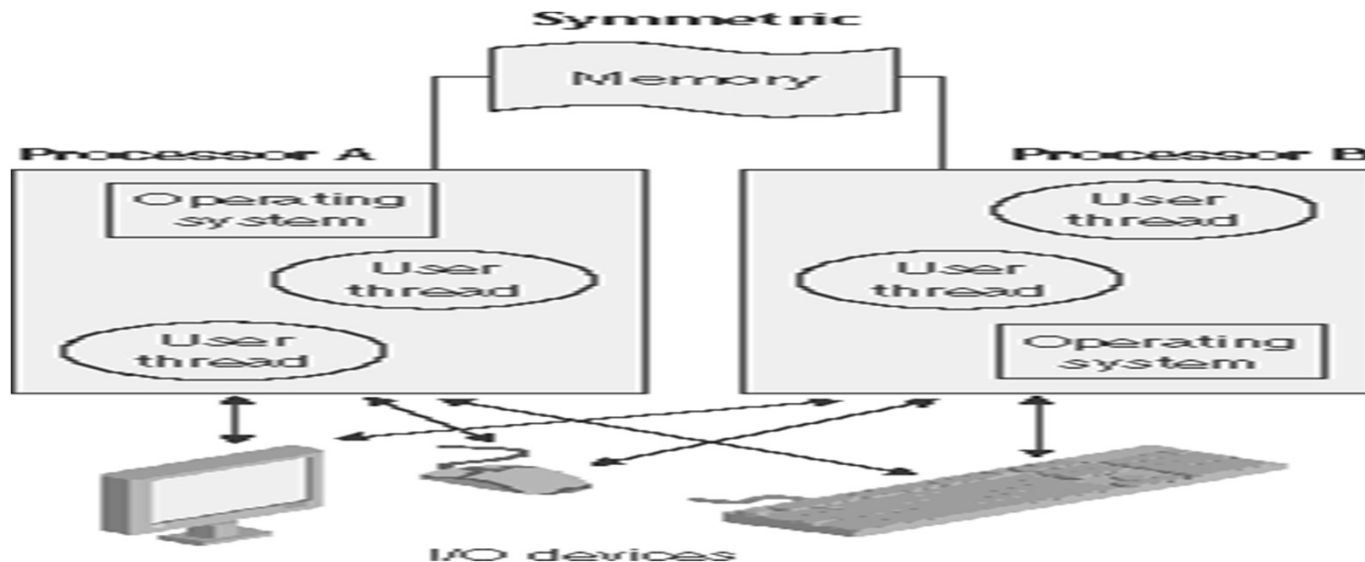
Symmetric Multiprocessing

- Symmetric multiprocessing (SMP) technology is used to get higher levels of performance.
- In symmetric multiprocessing, **any processor can run any type of thread**. The processors communicate with each other through shared memory.
- SMP systems provide better load-balancing and fault tolerance.



Symmetric Multiprocessing

- Because the operating system threads can run on any processor, the chance of hitting a CPU bottleneck is greatly reduced.
- All processors are allowed to run a mixture of application and operating system code.
- A processor failure in the SMP model only reduces the computing capacity of the system.



Difference Between Asymmetric and Symmetric Multiprocessing:

ASYMMETRIC MULTIPROCESSING	SYMMETRIC MULTIPROCESSING
In asymmetric multiprocessing, the processors are not treated equally.	In symmetric multiprocessing, all the processors are treated equally.
Tasks of the operating system are done by master processor.	Tasks of the operating system are done individual processor
No Communication between Processors as they are controlled by the master processor.	All processors communicate with another processor by a shared memory.
In asymmetric multiprocessing, process are master-slave.	In symmetric multiprocessing, the process is taken from the ready queue.
Asymmetric multiprocessing systems are cheaper.	Symmetric multiprocessing systems are costlier.
Asymmetric multiprocessing systems are easier to design	Symmetric multiprocessing systems are complex to design

Multiprocessor Systems

- **Tightly coupled System**
- More than one processor in close communication
- **Sharing the computer bus, the clock and sometimes memory and peripheral devices.**

Distributed Systems

- **Loosely Coupled Systems**
- Collection of processors
- **Do not share memory or a clock.**
- **Each processor has its own local memory.**
- The processors communicate with one another through various communication lines such as high speed buses or telephone lines.

Distributed Systems

- Collection of separate, possibly heterogeneous, systems networked together
 - **Network** is a communications path,
 - **TCP/IP** most common network protocol
 - Most OS support TCP/IP including the Windows and UNIX OS.

Distributed Systems

- Networks are typecast based on the distances between their nodes
 - **Personal Area Network (PAN)**
 - **Local Area Network (LAN)**
 - **Metropolitan Area Network (MAN)**
 - **Wide Area Network (WAN)**

Distributed Systems

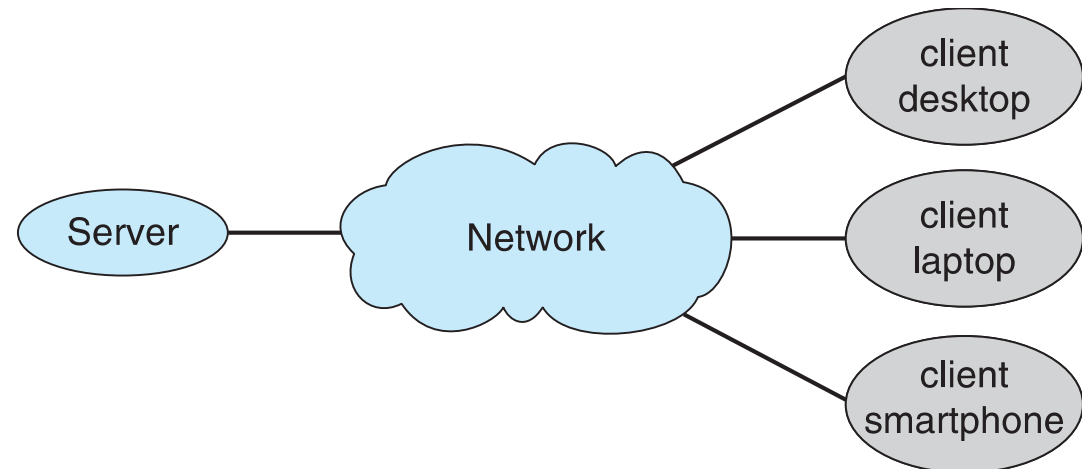
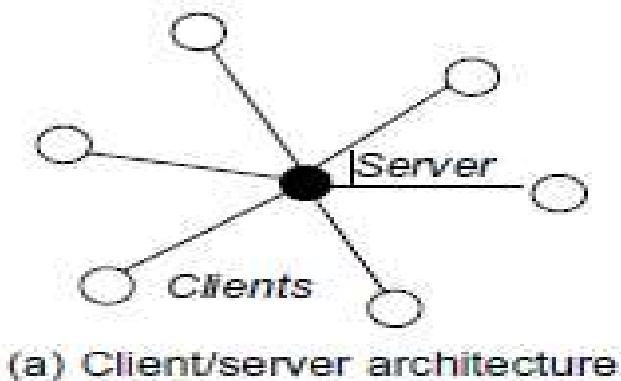
- Networks are typecast based on the distances between their nodes
 - **Personal Area Network (PAN)**-Short distance of several feet like Bluetooth Devices
 - **Local Area Network (LAN)**- within a room, a floor, or a building
 - **Metropolitan Area Network (MAN)**-link building within a city
 - **Wide Area Network (WAN)**- exists between buildings, cities or countries

Distributed Systems

- Types
 - Client-Server Systems
 - Peer-to-Peer Systems

Client-Server Systems

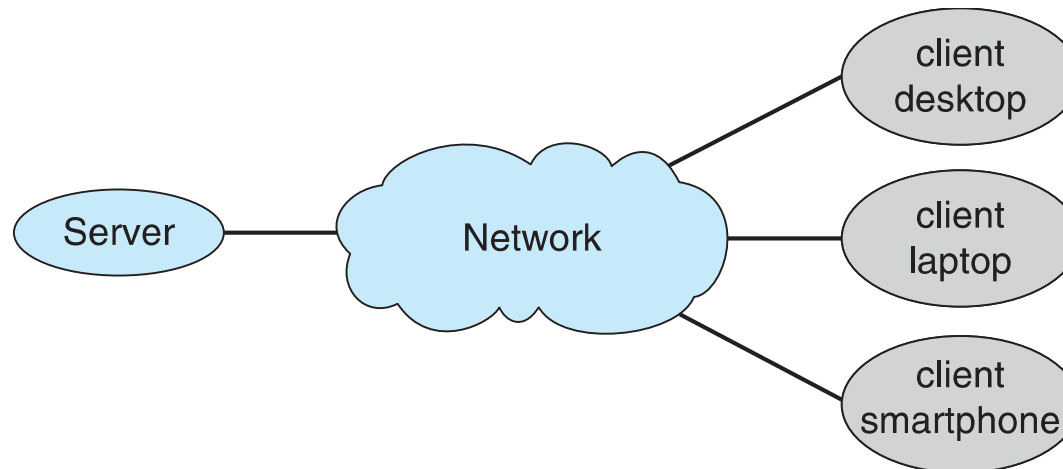
- Client-Server Computing
 - Centralized system act as server systems to satisfy requests generated by client systems
 - User interface handled by smart PCs.



Client-Server Systems

□ Server systems can be categorized as :

- ▶ **Compute-server system** provides an interface to client to request services (i.e., database), perform action, send back results to the client
- ▶ **File-server system** provides file system interface for clients to store and retrieve files i.e. create, update, read and delete files

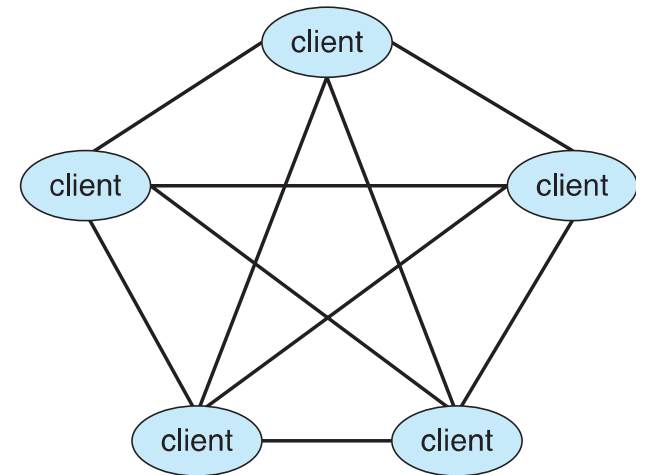


P2P Networks

- An alternative to the traditional client / server architecture where there typically is a single server and many clients.

Peer-to-Peer

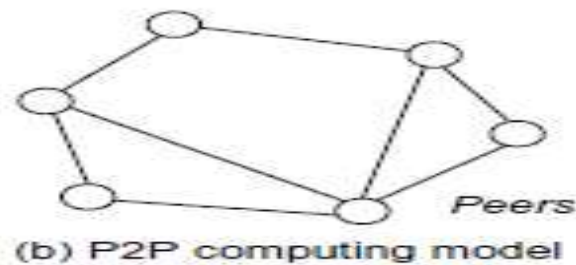
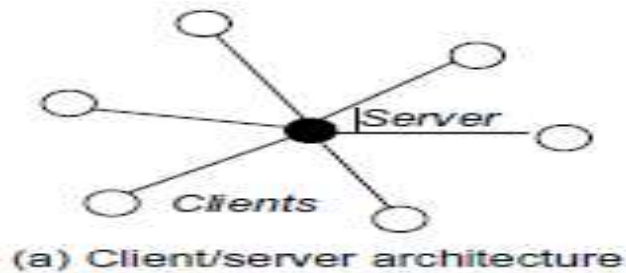
- Another model of distributed system
- P2P does not distinguish clients and servers
 - **All nodes are considered peers**



- **Why Peers?**

Peer-to-Peer

- Why Peers?
 - “A communications model in which each party has the same capabilities and
 - All play equal roles.
- Each node may act as client, server or both
- No peer has global view of the entire system.



P2P Networks

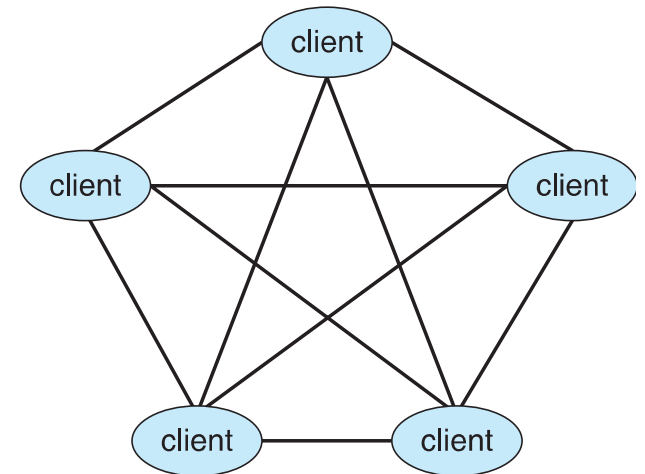
- Peers-
 - **Exchange resources with one another directly.**
 - Either party can initiate a communication session”

P2P Networks

- P2P Networks are autonomous and self-organised with free participation from peers.
- A P2P computing system is formed with peer hosts that are fully distributed without using a centralized server.

Peer-to-Peer

- Node must join P2P network
 - Registers its service with central lookup service on network, or
 - Broadcast request for service and respond to requests for service via ***discovery protocol***
- Examples such as Skype



P2P Networks

- Used widely in
 - Messaging,
 - Online Chatting,
 - Video streaming and
 - Social networking.

Peer-to-peer networking with BitTorrent

- BitTorrent is by far
 - the most popular peer-to-peer programs ever.

- ??

Peer-to-peer networking with BitTorrent

- BitTorrent is one of the most common protocols
 - for transferring large files,
 - such as **digital video files containing TV shows or video clips or digital audio files containing songs.**

BitTorrent Clients

- To send or receive files,
 - a person uses a BitTorrent client on their Internet-connected computer.
- A BitTorrent client is a
 - computer program that implements the BitTorrent protocol.
 - Popular clients include
 - µTorrent, Xunlei, Transmission, qBittorrent, Vuze, Deluge, BitComet and Tixati.

BitTorrent Clients

- FrostWire



- Tonido



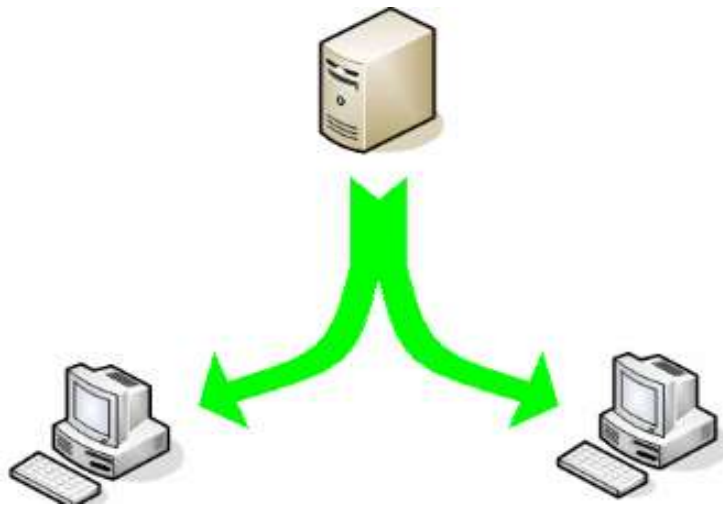
- μ Torrent



- Deluge



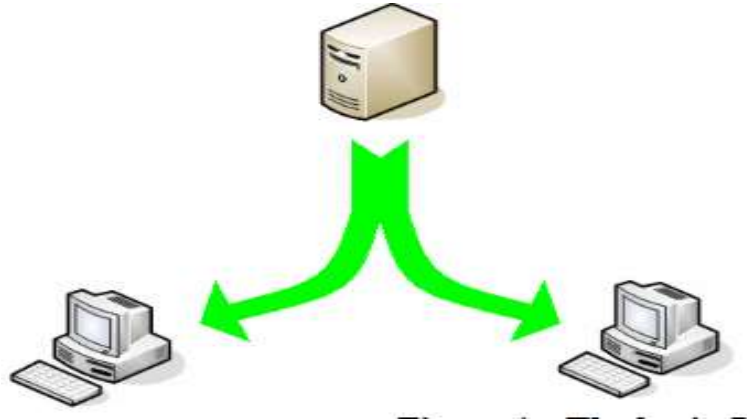
Typical Client Server



Downloading File from the server

P2P??

Typical Client Server



- In the figure, The time for the download to finish will be two times the time if only one peer was downloading from the server.

Basic Idea behind BitTorrent

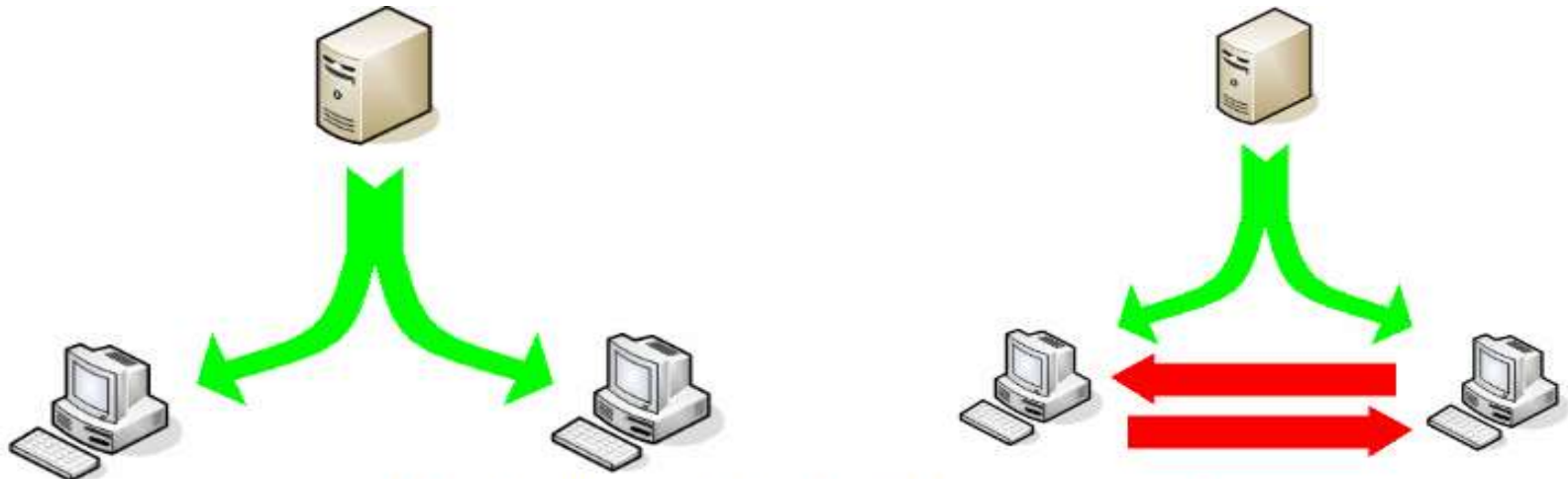


Figure 4 – The basic flow of the BitTorrent protocol.

Student Answersheets ??

Basic Idea behind BitTorrent

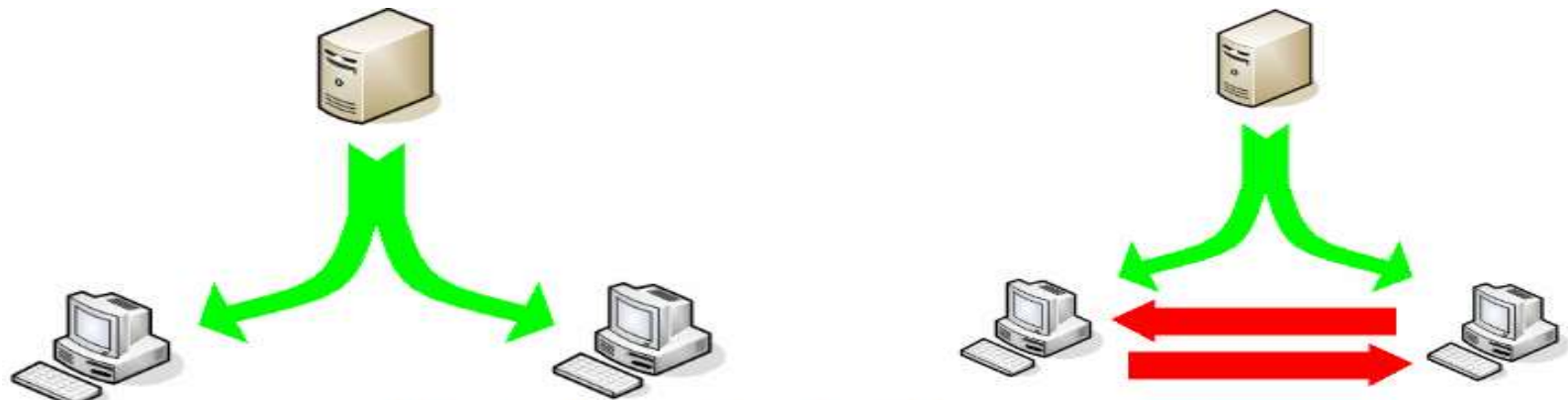
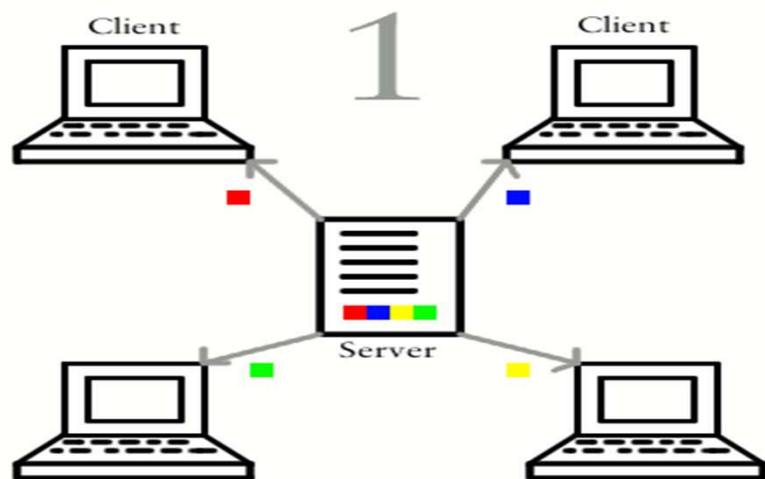
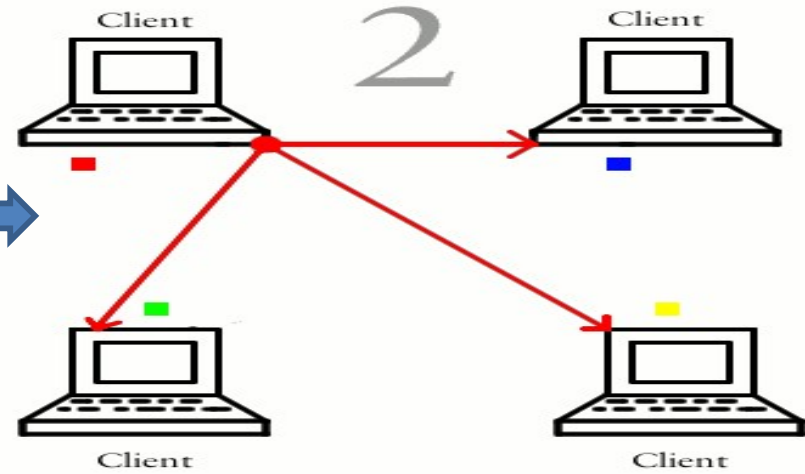


Figure 4 – The basic flow of the BitTorrent protocol.

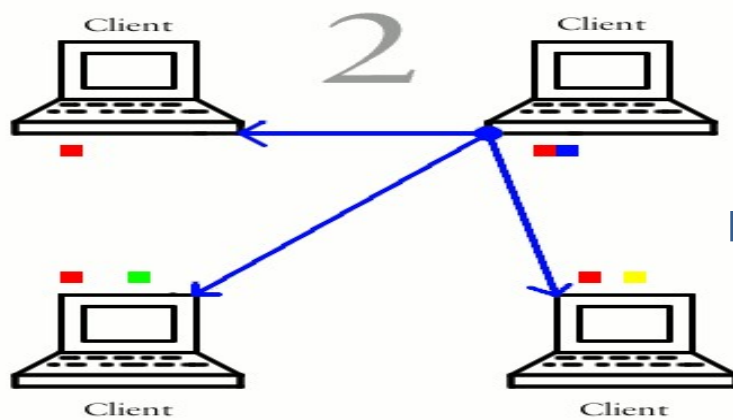
- The figure on the right shows an approach similar to BitTorrent.
 - By splitting the file
 - Send one part to each peer,
 - Let the peers download the part they are missing from each other,
 - Both download time and load on the server is reduced.
- Of course, the BitTorrent protocol is much more sophisticated than this simple example, but this shows the basic idea.



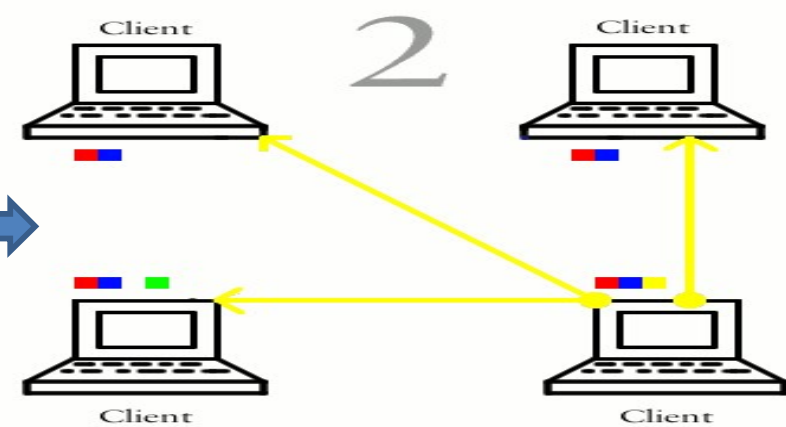
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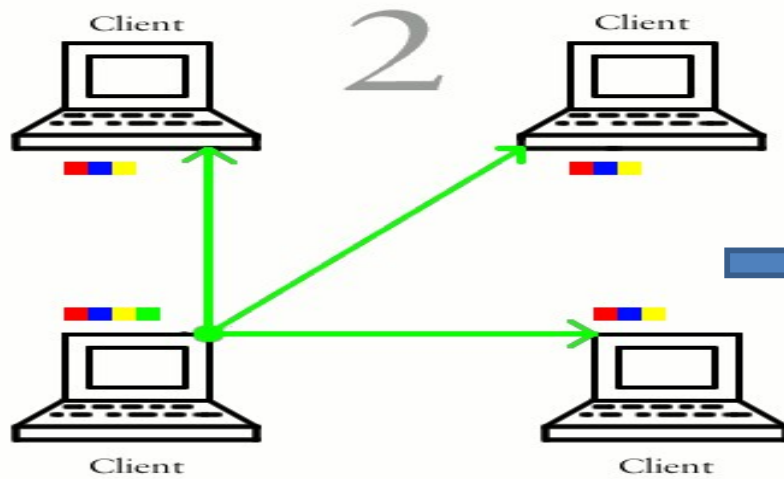
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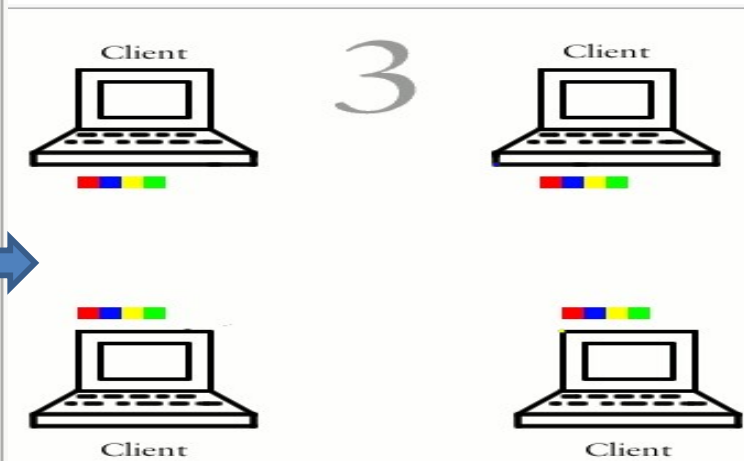
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Real Time Systems

- Special purpose OS
- **Used when rigid time requirements have been placed on the processing.**
- Often used as control device in Dedicated applications.

Real Time Systems

- Real-time operating systems (RTOS) are used in environments where **a large number of events, mostly external to the computer system, must be accepted and processed in a short time or within certain deadlines.**

Real Time Systems

Examples of the real-time operating systems:

- Airline traffic control systems,
- Command Control Systems,
- Industrial control,
- Telephone switching equipment,
- Flight control, and
- Real-time simulations.

Real Time Systems

- Real Time Systems **functions correctly only if it returns the correct result within its time constraints.**
- In Contrast-
 - Time Sharing Systems, **where it is desirable but not mandatory to respond quickly.**
 - Batch System, which may have **no time constraints at all.**

Real Time Systems

Two types of Real-Time Operating System which are as follows:

- Hard Real-Time Systems
- Soft Real-Time Systems

Hard Real-Time Systems:

- These OSs are meant for the applications where time constraints are very strict and even the shortest possible delay is not acceptable.
- These systems are built for saving life like automatic parachutes or air bags which are required to be readily available in case of any accident.

Soft Real-Time Systems:

- These OSs are for applications where for time-constraint is less strict.

Hard Real-Time Systems:

- This type of system can never miss its deadline.
- Missing the deadline may have disastrous consequences.

Soft Real-Time Systems:

- This type of system can miss its deadline occasionally with some acceptably low probability.
- Missing the deadline have no disastrous consequences.

Hard Real-Time Systems:

- The usefulness of result produced by a hard real time system decreases abruptly **if tardiness increases.**

Examples:

- **Satellite launch system**
- **Railway signalling system**

Soft Real-Time Systems:

- The usefulness of result produced by a soft real time system decreases gradually with increase in tardiness.

Examples:

- **Telephone switches.**
- **Multimedia Transmission and Reception**
- **Computer Games**
- **DVD Player**

Tardiness - **How late a real time system completes** its task with respect to its deadline.

HARD REAL TIME SYSTEM

In hard real time system, the size of data file is small or medium.

In this system response time is in millisecond.

Peak load performance should be predictable.

In this system safety is critical.

A hard real time system is very restrictive.

In case of an error in a hard real time system, the computation is rolled back.

Satellite launch, Railway signaling system etc.

SOFT REAL TIME SYSTEM

In soft real time system, the size of data file is large.

In this system response time are higher.

In soft real time system, peak load can be tolerated.

In this system safety is not critical.

A Soft real time system is less restrictive.

In case of an soft real time system, computation is rolled back to previously established a checkpoint.

DVD player, telephone switches, electronic games etc.

Match the following:

List – I	List – II
(a) Spooling	(i) Allows several jobs in memory to improve CPU utilization
(b) Multiprogramming	(ii) Access to shared resources among geographically dispersed computers in a transparent way
(c) Time sharing	(iii) Overlapping I/O and computations
(d) Distributed computing	(iv) Allows many users to share a computer simultaneously by switching processor frequently

codes:

	(a)	(b)	(c)	(d)
(1)	(iii)	(i)	(ii)	(iv)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(ii)	(iii)	(iv)	(i)

(A) (1)

(B) (2)

(C) (3)

(D) (4)

Match the following:

List – I	List – II
(a) Spooling	(i) Allows several jobs in memory to improve CPU utilization
(b) Multiprogramming	(ii) Access to shared resources among geographically dispersed computers in a transparent way
(c) Time sharing	(iii) Overlapping I/O and computations
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codes:

	(a)	(b)	(c)	(d)
(1)	(iii)	(i)	(ii)	(iv)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(ii)	(iii)	(iv)	(i)

- (A) (1)
(B) (2)
(C) (3)
(D) (4)

Answer: (B)

Explanation:

- Spooling provides Overlapping I/O and computations.
- Multiprogramming allows several jobs in memory to improve CPU utilization.
- Time sharing allows many users to share a computer simultaneously by switching processor frequently.
- Distributed computing accesses to shared resources among geographically dispersed computers in a transparent way.

So, option (B) is correct.

SPOOL is an acronym for **simultaneous peripheral operations on-line**.

- A large buffer from hard disk
 - Buffer can store the data through I/O
 - Because I/O is slow and CPU is fast.
- In a computer system peripheral equipments, such as printers and punch card readers etc (batch processing), are very slow relative to the performance of the rest of the system.

SPOOL is an acronym for **simultaneous peripheral operations on-line**.

- In computing, spooling is a specialized form of multi-programming for the purpose of copying data between different devices
-
- It is usually used for mediating between a computer application and a slow peripheral, such as a printer.
- A dedicated program, the **spooler**, maintains an orderly sequence of jobs for the peripheral and feeds it data at its own rate.

ISRO | ISRO CS 2017 – May | Question 71

Which of the following statement is true?

- (A)** Hard real time OS has less jitter than soft real time OS
- (B)** Hard real time OS has more jitter than soft real time OS
- (C)** Hard real time OS has equal jitter as soft real time OS
- (D)** None of the above

ISRO | ISRO CS 2017 – May | Question 71

Which of the following statement is true?

- (A) Hard real time OS has less jitter than soft real time OS
- (B) Hard real time OS has more jitter than soft real time OS
- (C) Hard real time OS has equal jitter as soft real time OS
- (D) None of the above

Answer: (A)

Explanation: Jitter is the variation / displacement between the signals or data being sent.

Hard real operating systems deal with more sensitive systems which require strict time deadlines like engine control systems, satellite launching systems etc while soft real operating systems do not require strict timing constraints and a bit delay is permissible, like mobile phones, online database systems. So hard real operating systems require minimised jitter.

ISRO | ISRO CS 2008 | Question 51

What is the name of the operating system that reads and reacts in terms of actual time.

- A. Batch system
- B. Quick response system
- C. Real time system
- D. Time sharing system

ISRO | ISRO CS 2008 | Question 51

What is the name of the operating system that reads and reacts in terms of actual time.

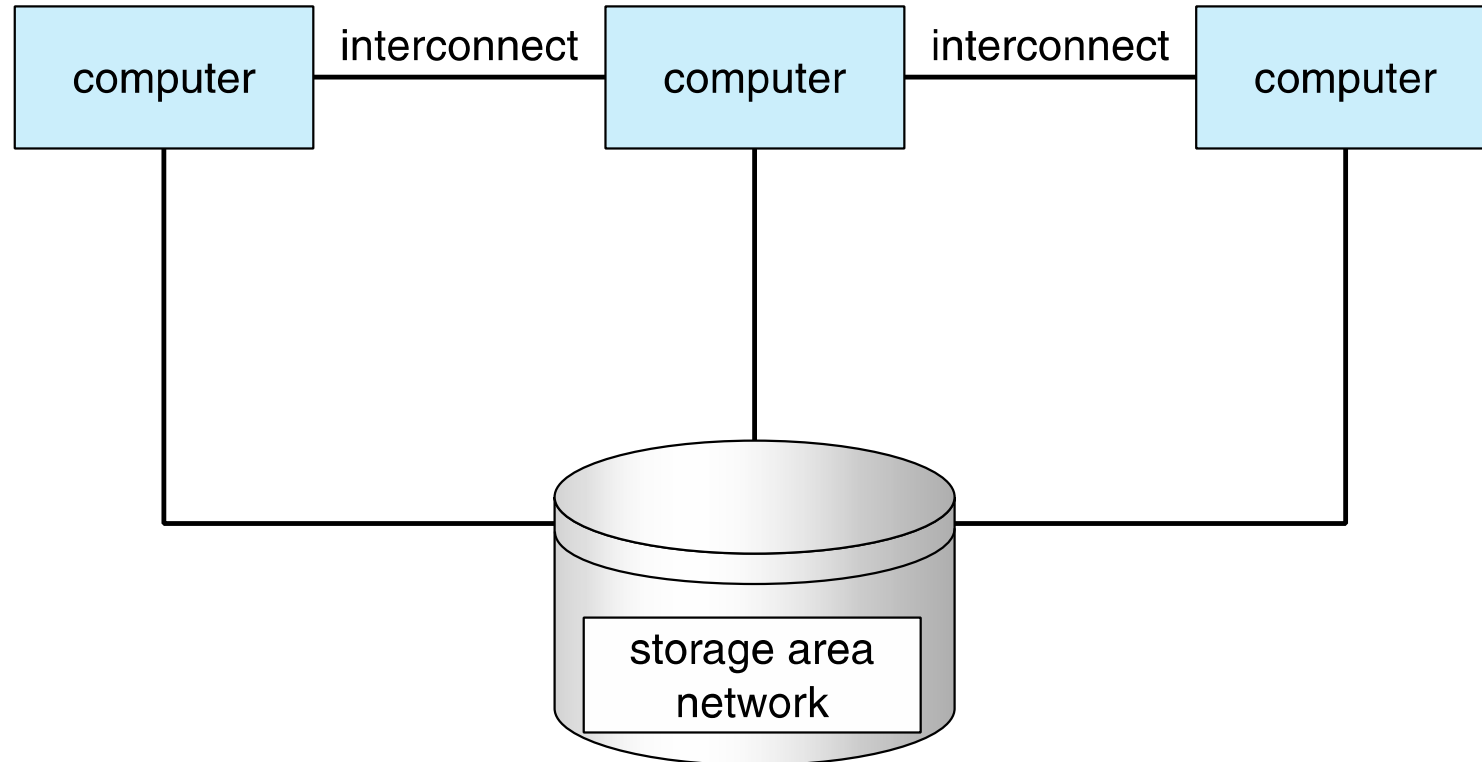
- A. Batch system
- B. Quick response system
- C. Real time system
- D. Time sharing system

Answer: (C)

Explanation: A real-time operating system is an operating system that guarantees to process events or data by a specific moment in time.

Option (C) is correct.

Clustered Systems



- Like multiprocessor systems, but multiple systems working together Usually sharing storage via a **storage-area network (SAN)**

Clustered Systems

- Provides a **high-availability** service which survives failures
- Clusters are used for **high-performance computing (HPC)**
 - Applications must be written to use **parallelization**
- Some have **distributed lock manager (DLM)** to avoid conflicting operations

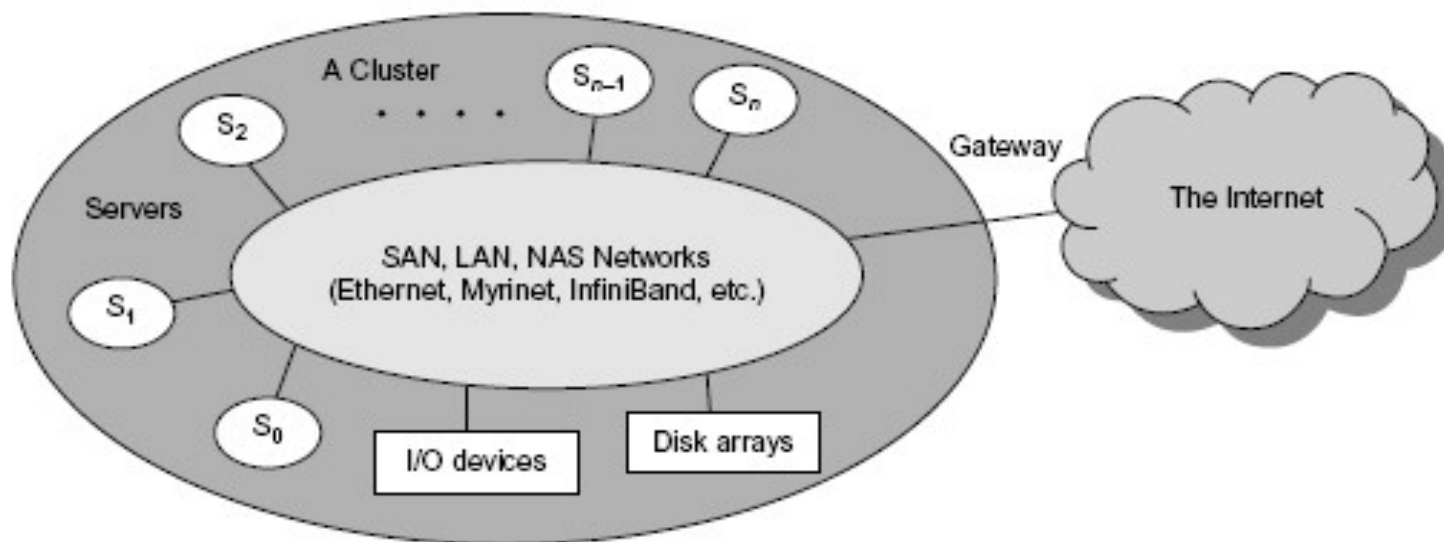
Cluster for Massive Parallelism

- A collection of
 - interconnected stand alone computers
 - which can work together collectively and
 - cooperatively as a single integrated computing resource pool.
- Deployed to
 - improve speed and/or reliability over that provided by a single computer, while
 - being much more cost effective than single computer of comparable speed or reliability

Cluster Architecture

Through hierarchical construction using a SAN, LAN or WAN, one can build scalable clusters with an increasing number of nodes.

The cluster is connected to the internet via a virtual private network(VPN) gateway.



Single System Image

- An ideal cluster should merge multiple system images into a single system image (SSI).
- Cluster designers desire a cluster operating system to support SSI at various levels, including the sharing of CPUs, memory and I/O across all cluster nodes.
- SSI is an illusion created by software or hardware that presents a collection of resources as one integrated, powerful resource.
- SSI makes the cluster appear like a single machine to the user.

SSI

- Suppose a workstation has:
 - A 300 Mflops/second processor
 - 512 MB of memory
 - 4 GB disk
 - Support 50 active users and 1,000 processes.
- By clustering 100 such workstations, we should get a single system that is equivalent to one huge workstation, or a megastation, that has:
 - 30 Gflops/second processor
 - 50 GB of memory
 - 400 GB disk
 - Support 5,000 active users and 100,000 processes
- This is an appealing goal, but it is very difficult to achieve.
- SSI techniques are aimed at achieving this goal.

- **Asymmetric clustering** has one machine in hot-standby mode
- **Symmetric clustering** has multiple nodes running applications, monitoring each other

Clustered Systems

- Asymmetric clustering-
 - One m/c is in hot standby mode while the other (primary node) is running applications.
 - The hot standby host does nothing but monitor the active server.
 - If that server fails, the hot standby host becomes the active server.
 - The standby node is powered on (hot) and running some monitoring programs to communicate heartbeat signals to check the status of the primary node, but is not actively running other useful workloads.

Clustered Systems

- Symmetric clustering-
 - Two or more hosts are running applications and they are monitoring each other.
 - More efficient, as it uses all of the available hardware.